

Natural Language to SQL query Interface using Gemini & Streamlit

An Application for Interactive Data Analysis presented by Mandakini Srivastava on April 22, 2025.

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Introduction to Data Insights App



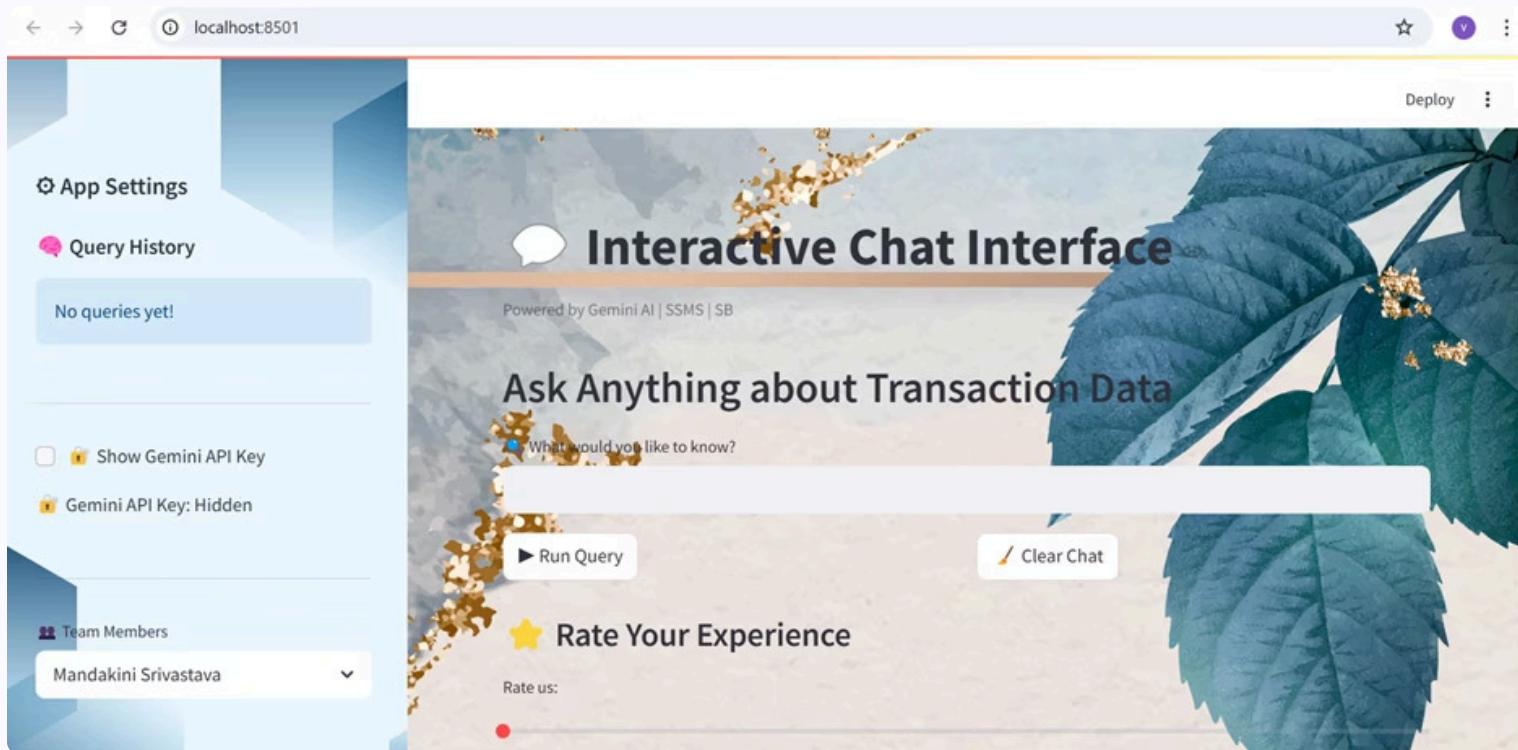
App Overview

Intuitive interface for database table analysis.



Purpose

Enable natural language queries for data exploration.



Features of the Application

Connected Database Selection

Easy table selection for queries.

Natural Language Questions

User-friendly plain English queries.

Auto SQL Query Generation

Powered by advanced language models.

SQLQuery4.sql - localhost\SQLEXPRESS.test-manu (MANU-2786\DELL (52)) - Microsoft SQL Server Management Studio

File Edit View Query Project Tools Window Help

test-manu Execute

Object Explorer

Connect

localhost\SQLEXPRESS (SQL Server 16.0.1000 - MANU-2786\DELL)

- Databases
 - System Databases
 - Database Snapshots
 - test-manu
 - Database Diagrams
 - Tables
 - System Tables
 - FileTables
 - External Tables
 - Graph Tables
 - dbo.CustomerTable
 - dbo.SalesTable
 - dbo.transaction_data
 - Columns
 - Keys
 - Constraints
 - Triggers
 - Indexes
 - Statistics
 - dbo.TransactionLog
 - Views
 - External Resources
 - Synonyms
 - Programmability

SQLQuery4.sql - lo...NU-2786\DELL (52)

```
SELECT TOP (1000) [Transaction ID]
, [Sender Account ID]
, [Receiver Account ID]
, [Transaction Amount]
, [Transaction Type]
, [Timestamp]
, [Transaction Status]
, [Fraud Flag]
, [Geolocation (Latitude Longitude)]
, [Device Used]
, [Network Slice ID]
, [Latency (ms)]
, [Slice Bandwidth (Mbps)]
, [PIN Code]
FROM [test-manu].[dbo].[transaction_data]
```

100 %

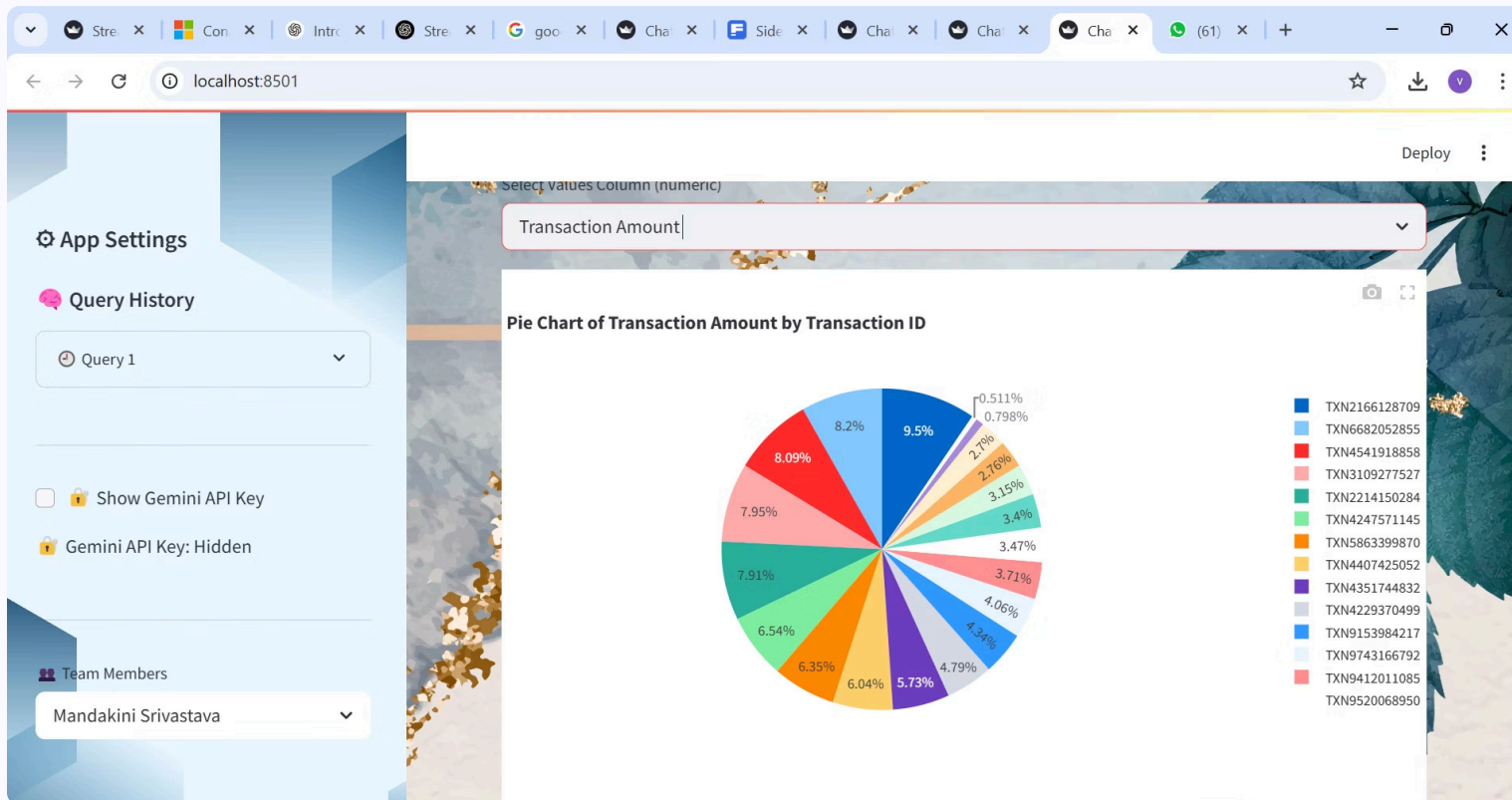
Results Messages

	Transaction ID	Sender Account ID	Receiver Account ID	Transaction Amount	Transaction Type	Timestamp	Transaction Status	Fraud Flag
1	TXN9520068950	ACC14994	ACC16656	495.9	Deposit	2025-01-17 10:14:00	Failed	True
2	TXN9412011085	ACC58958	ACC32826	529.62	Withdrawal	2025-01-17 10:51:00	Success	False
3	TXN4407425052	ACC56321	ACC92481	862.47	Withdrawal	2025-01-17 10:50:00	Failed	False
4	TXN2214150284	ACC48650	ACC76457	1129.88	Transfer	2025-01-17 10:56:00	Success	True
5	TXN4247571145	ACC60921	ACC11419	933.24	Deposit	2025-01-17 10:25:00	Success	True
6	TXN2515439857	ACC18381	ACC76710	449.56	Withdrawal	2025-01-17 10:01:00	Failed	False

Query executed successfully. localhost\SQLEXPRESS (16.0... MANU-2786\DELL (52) test-manu 00:00:00 1,000 rows

Ready Ln 16 Col 1 Ch 1 INS

Visualization Options



Result Display

- Tables
- Bar Charts
- Line Charts
- Pie Charts

Instant Insights

Auto-generated data summaries for quick understanding.

User Interaction Enhancements

Dynamic Charting

Select chart type and axes dynamically.

Insight Summaries

Automated explanations of trends and key data points.





Installation Prerequisites

1. Install Python 3.12
2. Run: `pip install google.generativeai`



Starting the Application

Use the command:

```
streamlit run Interactive_Chat_Interface.py
```

Project Structure Overview

Interactive_Chat_Interface.py

Streamlit app main file

Api_Sql_Gemini.py

SQL logic implementation

README.md

Project documentation

data

CSV files storage



Database Configuration

Database:

Username:

Password:

Port:

Schema:

Environment Variables

API-KEY:

API-SECRET:

PORT-LOPEH:

PORT-TOKEN:

Save Configuration

Configuration Tips



Verify Database Connectivity

Ensure connection parameters are correct.



Set Environment Variables

Provide credentials securely for access.

Conclusion and Next Steps

- Simplifies data analysis through natural language.
- Delivers powerful, interactive analytics.
- Engages a broad range of users.

Next Steps: Deploy, gather feedback, and iterate on features for improved user experience and expanded data sources.

You may ask questions...

Thank you 🙏